Marginally Useful: Conformal Prediction Is Early Stopping at an Information Price

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Abstract

We view conformal prediction as a repairable univariate transform to be included alongside other time-series operators in a forecasting grammar. By this device we evaluate the wisdom of stopping after the conformalizing step, something that is tacit in conformal prediction and departed from by only two of the 623 papers proposing conformal methods in our survey. Across 17,364 economic time-series, early stopping costs approximately 0.4 nats. We interpret this through the lens of a new information identity we term the conformal information gap, whereby it is possible to formalize why conformal calibration, at a fixed score and stratification, cannot exploit forecastability.

1 Introduction

The most any forecast of an outcome Y can gain over the unconditional distribution of Y, in expected log score, is the mutual information $\mathrm{I}(Y;X)$ between the outcome and the information X available to the forecaster (DelSole, 2004). Catt (2026) takes this quantity as the definition of the forecastability of Y, and calls the gain a method actually achieves its exploitability. We ask how much forecastability a conformal forecaster exploits, and we will prove that, once the score and the stratification are fixed, conformal calibration exploits none of it.

The conformal recipe is familiar. We fit a base model and then, on n held-out points, compute a scalar error for each (for instance the absolute residual). We then sort the errors, take the $\lceil (n+1)\,0.9 \rceil$-th smallest, and quote the model's prediction plus or minus that value. If the joint law of the held-out data and the future point is unchanged by permuting them, the quoted interval contains the truth at least 90% of the time, at any sample size, with no further assumption on the model or the data distribution.

The last step of the recipe applies the empirical distribution function of the held-out errors, and the quoted interval is read off its quantiles. Nothing is fitted to the output of that step. The recipe stops there, and that is the reason we choose to describe conformal prediction as a form of early stopping. The stop is easy to miss because conformal prediction is usually presented as a self-contained procedure.

Once the empirical map is repaired, however, it is an invertible univariate transform of the error (Section 2). It then belongs in a grammar of similar transforms, alongside subtracting a running mean, dividing by a running scale, differencing, serial whitening, and the normal quantile

transform of hydrological forecasting (Bogner et al., 2012). A forecaster working in that grammar can apply the transform and keep modelling its output. As far as we know the literature does not describe conformal prediction as early stopping, and no paper justifies stopping at the empirical map. Of 623 papers proposing conformal methods, only two go past the map, and both only to size a multivariate set (Appendix A). We show that the stop carries a steep empirical price.

We make two contributions. We introduce and prove the conformal information gap identity (Proposition 2), which fixes, exactly and at a fixed calibration sample, the part of the expected log regret that no calibration choice can reach. We then measure what stopping costs on economic series, and how much of that cost composing a model past the pooled law recovers (Section 4).

2 Conformal prediction as a univariate transform

Viewed as a univariate transform, conformal prediction is in need of repair. Its raw form is a step function, and once repaired it is an invertible map that can sit anywhere in a chain of transforms.

Let $R_1, \dots, R_n$ be calibration scores, for example held-out residuals. Split conformal prediction applies the fence-post empirical map

$$E_n(r) \;=\; \frac{1 + \#\{i \le n : \; R_i \le r\}}{n+1} \tag{1}$$

and reads the prediction set off its level sets (Vovk et al., 2005; Lei et al., 2018). The value of E_n at a new score is a conformal p-value, and E_n itself is the unrandomized conformal predictive distribution of Vovk et al. (2019). If $R_1, \dots, R_{n+1}$ are exchangeable and almost surely distinct, the rank of R_{n+1} among all $n+1$ scores is uniform, so $E_n(R_{n+1})$ is uniform on the grid $\{k/(n+1) : k = 1, \dots, n+1\}$. That is the coverage guarantee.

A *transform*, in the sense of Cotton (2026a), is an online map with a state. Its forward function consumes an observation and the current state and emits a transformed observation. Its inverse maps a predictive distribution for the transformed observation back to one for the original. Given the state, the forward function is a strictly increasing change of variables, so predictive densities pull back exactly, Jacobians included. A forecaster is a *chain* of transforms followed by a *leaf*, a predictive law for whatever the last transform emits.

Table 1 lists the transforms of the `skaters` library in which that grammar is implemented. Every one of them is invertible given its state, and a chain may stack any of them in any order.

The map (1) is a step function. It has no inverse and no density, so it is not a transform. Cotton (2026b) repairs it by interpolation. Let $r_{(1)} < \cdots < r_{(n)}$ be the order statistics of the scores, give the i-th knot the probability $i/(n+1)$, which is the left limit of E_n at $r_{(i)}$ and the exact probability that a new exchangeable score falls at or below $r_{(i)}$, and set $z_{(i)} = \Phi^{-1}\big(i/(n+1)\big)$.

The repaired map m_n is any strictly increasing bijection of the real line with $m_n(r_{(i)}) = z_{(i)}$ for $i = 1, \dots, n$, rebuilt as new scores arrive. Appendix B gives the interpolation and tails used here. The repair keeps the certificate at the knots, $\Pr\{m_n(R_{n+1}) \le z_{(i)}\} = i/(n+1)$, and $\Phi(m_n(R_{n+1}))$ is within Kolmogorov distance $1/(n+1)$ of uniform whatever the interpolation (Appendix B).

With m_n in hand the two positions are precise. A forecaster *stops* if m_n is the last transform in its chain and the leaf is the fixed standard normal law. Its predictive is then $N(0,1)$ pulled back through m_n, which is the smoothed empirical law of the calibration scores: the conformal predictive, and nothing downstream of it. A forecaster *composes* if it keeps m_n and fits further transforms or a leaf to the stream $z_t = m_n(R_t)$, then pulls the result back through m_n. Composing

transform	forward	inverse of a predictive D	removes
`ema`(α)	$y'_t = y_t - \ell_t$	$D \mapsto D + \ell_t$	level
`difference`	$y'_t = y_t - y_{t-1}$	cumulative sum, variance $\sum_h \sigma_h^2$	random walk
`drift`(α, λ)	$y'_t = \Delta y_t - \hat{\mu}_t$	$y_t + h\hat{\mu}_t + \sum \varepsilon$	random walk with drift
`holt_linear`(α, β)	$y'_t = y_t - (\ell_t + b_t)$	$\ell_t + hb_t + \varepsilon$	level and trend
`ar`(p)	$y'_t = y_t - \sum_j \hat{\phi}_j y_{t-j}$	AR reconstruction, variance propagated	autoregression (online least squares)
`grouped_ar`(L)	as `ar`, coefficients grouped by lag	as `ar`	long lags
`fractional_difference`(d)	$y'_t = (1-B)^d y_t$	$(1-B)^{-d}$	long memory
`standardize`(α)	$y'_t = (y_t - \hat{\mu}_t)/\hat{\sigma}_t$	$D \mapsto \hat{\sigma}_t D + \hat{\mu}_t$	level and scale
`garch`(ω, α, β)	$y'_t = y_t/\hat{\sigma}_t$	$D \mapsto \hat{\sigma}_t D$	volatility clustering
`seasonal_difference`(s)	$y'_t = y_t - y_{t-s}$	shift by the lagged value	periodicity
`power_transform`(p)	$y'_t = \mathrm{sign}(y_t)\lvert y_t\rvert^p$	exact pushforward	heavy tails
`theta`(α)	$y'_t = y_t - \mathrm{SES}_t$	shift by smoothed level and drift	theta method
`yeo_johnson`(λ)	signed Box–Cox with parameter λ	exact pushforward	coordinates
`ou`(κ)	deviation from a running mean	exact Ornstein–Uhlenbeck moments	mean reversion
repaired conformal map	$y'_t = m_n(y_t)$	pull back through m_n^{-1}, Jacobian included	shape of the pooled law

Table 1: Transforms of the forecasting grammar, from the `skaters` library (Cotton, 2026a). Here ℓ_t, b_t, $\hat{\mu}_t$ and $\hat{\sigma}_t$ are running level, slope, mean and scale estimates held in the transform's state, h is the forecast horizon and B the backshift operator. The last row is the repaired empirical map of this section.

keeps the certificate at the position of m_n. It does not carry the certificate to the final predictive, which is the price of continuing.

3 The conformal information gap

A *nonconformity score* is a map $A : \mathcal{X} \times \mathbb{R} \to \mathbb{R}$ assigning a scalar to each candidate outcome y at input x. A *stratification* $\kappa : \mathcal{X} \to \mathcal{K}$ assigns each input to a category. Both are fit on a training fold and frozen, since exchangeability of calibration and test scores is what the coverage argument uses. The schemes covered by the proposition are those whose score is, for each x, a strictly increasing absolutely continuous bijection in y: signed-residual and normalized signed-residual schemes, and predictive systems built on invertible score transformations.

Folded scores such as the absolute residual $|y - \widehat{\mu}(x)|$ and quantile-regression scores are not bijections in y. Remark 7 treats the absolute residual in its symmetric reading. Quantile-regression scores need a separate treatment. The empirical arms in Section 4 are of the covered type.

Such a scheme emits the pooled empirical law of calibration scores in the test point's stratum,

pushed back through $A(x,\cdot)^{-1}$. For log scoring the emitted law must have a density, so let $\widehat{G}_\kappa$ be an absolutely continuous estimate of the pooled score law (a kernel-smoothed or interpolated empirical law). The raw empirical measure is atomic and would score $-\infty$. Write $W = A(X,Y)$, let G_κ be the population pooled law, and let the benchmark be the conditional oracle, whose expected negative log score is the conditional entropy $H(Y \mid X)$ when that is finite. We state results as expected log regret $\mathbb{E}\log\{p(Y \mid X)/\widehat{f}_X(Y)\}$, which stays finite when the two log losses do not, because the Jacobian of the score cancels inside the ratio. Smaller is better.

The proof of the floor uses one standard identity, the chain rule for relative entropy along a product reference (Cover and Thomas, 2006). In coding theory it splits redundancy into a dependence term and a mismatch term (Clarke and Barron, 1990; Grünwald, 2007).

Lemma 1 (Chain rule for relative entropy)**.** *Let (U,V) have a joint density with $\mathrm{I}(U;V) < \infty$, let p_V be the marginal density of V, and let q be any density with $\mathrm{KL}(P_V \,\|\, Q) < \infty$. Then*

$$\mathbb{E}\log\frac{p(V \mid U)}{q(V)} \;=\; \mathrm{I}(U;V) + \mathrm{KL}\big(P_V \,\|\, Q\big).$$

Proof. Insert the marginal density p_V inside the logarithm:

$$\mathbb{E}\log\frac{p(V \mid U)}{q(V)} \;=\; \mathbb{E}\log\frac{p(V \mid U)}{p_V(V)} \;+\; \mathbb{E}\log\frac{p_V(V)}{q(V)}.$$

The first expectation is $\mathrm{I}(U;V)$ by definition. In the second the integrand depends on V alone, and V has law P_V, so it equals

$$\int p_V(v)\log\frac{p_V(v)}{q(v)}\,dv \;=\; \mathrm{KL}\big(P_V \,\|\, Q\big).$$

Both terms are finite under the assumptions. □

The floor needs two further facts about conformal schemes. The first is that the Jacobian of an invertible score cancels between truth and forecast, so the input reaches the predictive only through the frozen score and stratum. The second is that every calibration operation acts on the pooled scores alone. Together they give the following result, which to our knowledge has not been stated before.

Proposition 2 (Conformal information gap identity)**.** *Fix the training fold and the calibration sample, so $(A, \kappa, \widehat{G}_\kappa)$ are deterministic and the emitted predictive depends on the test input only through $A(x,\cdot)$ and $\kappa(x)$, let (X,Y) be a test pair independent of both, and let $W = A(X,Y)$ have conditional density $p_W(\cdot \mid x)$ given $X = x$ and pooled density g_κ given the stratum. Assume $\mathbb{E}|\log p_W(W \mid X)|$, $\mathbb{E}|\log g_{\kappa(X)}(W)|$ and $\mathbb{E}|\log \widehat{g}_{\kappa(X)}(W)|$ are finite, so that every term below is finite. Then the expected log regret of the conformal predictor against the conditional oracle decomposes exactly as*

$$\mathbb{E}\log\frac{p(Y \mid X)}{\widehat{f}_X(Y)} \;=\; \mathrm{I}(W; X \mid \kappa) \;+\; \mathbb{E}_\kappa\,\mathrm{KL}\big(G_\kappa \,\|\, \widehat{G}_\kappa\big),$$

where $\mathrm{I}(W; X \mid \kappa)$ is the conditional mutual information between score and input given the stratum. We call it the conformal information gap, or the floor. Unconditionally, take the outer expectation over the calibration sample; the first term is unchanged.

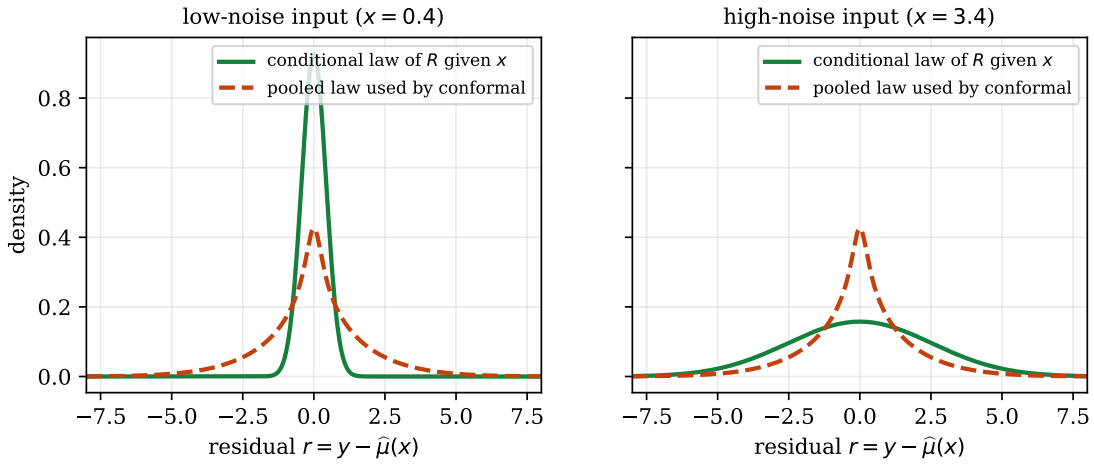


Figure 1: The conformal information gap when the error spread grows with the input. A single-stratum conformal scheme uses the pooled law (dashed) for every input. It is too wide for low-noise inputs and too narrow for high-noise ones. Its expected log regret over the conditional law (solid) is $\mathrm{I}(R; X)$, here 0.243 nats per observation.

The second term is nonnegative and is the only part any calibration choice, estimator, quantile convention, randomization, or level, can affect. At a fixed calibration sample it vanishes if and only if $\widehat{G}_\kappa = G_\kappa$ almost everywhere. KL-consistent estimation of the pooled law with growing calibration mass per stratum drives it to zero asymptotically, which is an assumption on the estimator, not a consequence of the setup.

The proof is a change of variables in which the Jacobian of A cancels between truth and forecast, then Lemma 1 within each stratum (Appendix C). The input reaches the predictive only through the frozen score and the frozen stratum. Whatever information about the score the input still carries within a stratum is therefore lost before calibration begins, and more calibration data does not recover it. With a location score and one stratum the floor is $\mathrm{I}(R; X)$, the mutual information between the model's error and its input.

Figure 1 draws this case when the spread of the error grows with the input. The pooled law is too wide where the noise is low and too narrow where it is high. With X uniform on $[0, 4]$ and error standard deviation $0.15 + 0.7x$, the floor is $\mathrm{I}(R; X) = 0.243$ nats per observation, and no calibration choice reduces it.

Remark 3 (When the floor is zero). The floor is zero exactly when the frozen pair renders the score independent of the input within strata: $\mathrm{I}(W; X \mid \kappa) = 0$ iff $W \perp X \mid \kappa$. When X carries the full information set available downstream, that is the statement that the modelling was finished before the conformal step. Conformal calibration is especially valuable when one is unwilling to assume that. Its guarantee does not diagnose whether the floor is zero, however, and one may deploy it on perfectly finished residuals purely for finite-sample validity. When the floor is positive, no calibration choice recovers any of it.

Lemma 4 (Convergence of the pooled law)**.** *Let the calibration pairs be i.i.d. copies of (X, Y), or a stationary ergodic sequence with that marginal law, let there be finitely many strata, each of positive probability, and let $\widehat{G}_{\kappa,n}$ be the empirical distribution of the scores in stratum κ. Then*

$\sup_w |\widehat{G}_{\kappa,n}(w) - G_\kappa(w)| \to 0$ *almost surely. If the smoothed estimates satisfy* $\mathrm{KL}(G_\kappa \,\|\, \widehat{G}_{\kappa,n}) \to 0$ *in every stratum, the expected log regret tends to the floor* $\mathrm{I}(W; X \mid \kappa)$.

Proof. In the i.i.d. case the uniform convergence is the Glivenko–Cantelli theorem. In the stationary ergodic case the scores $W_i = A(X_i, Y_i)$ are themselves stationary and ergodic, so the ergodic theorem applied to the indicators $\mathbf{1}\{W_i \le w\}$ gives almost sure convergence at each w in a countable dense set. Because distribution functions are monotone and G_κ is continuous, convergence on a dense set extends to uniform convergence. The second statement follows from Proposition 2, whose second term tends to zero under KL-consistency. □

Exchangeability alone does not give Lemma 4. An infinite exchangeable sequence is a mixture of i.i.d. sequences, and its empirical law converges to that of the random directing measure. Coverage therefore needs only exchangeability and holds at every sample size. The log-score statement is exact for every frozen pooled law (Proposition 2) and asymptotic only in identifying the law a conformal scheme ends up with.

Following DelSole (2004) and Catt (2026), the *forecastability* of Y given X is

$$\mathrm{I}(Y; X) \;=\; \mathbb{E} \log p(Y \mid X) - \mathbb{E} \log p_Y(Y),$$

the most any forecast can gain over the unconditional density p_Y in expected log score. The *exploitability* of a forecast is the gain it actually achieves. It is at most the forecastability and can be negative.

Corollary 5 (Gain of a conformal scheme over the unconditional law)**.** *Under Proposition 2, let* p_Y *be the unconditional density of* Y *and suppose* $\mathrm{I}(Y; X) < \infty$*. The conformal predictive gains over the unconditional law*

$$\mathbb{E} \log \widehat{f}_X(Y) - \mathbb{E} \log p_Y(Y) \;=\; \mathrm{I}(Y; X) - \mathrm{I}(W; X \mid \kappa) - \mathbb{E}_\kappa \mathrm{KL}\big(G_\kappa \,\|\, \widehat{G}_\kappa\big).$$

The gain is at most $\mathrm{I}(Y; X) - \mathrm{I}(W; X \mid \kappa)$*, and the frozen score and stratification fix that bound. It is negative whenever* $\mathrm{I}(W; X \mid \kappa) > \mathrm{I}(Y; X)$.

Proof. Add and subtract the oracle's expected log score:

$$\mathbb{E} \log \widehat{f}_X(Y) - \mathbb{E} \log p_Y(Y) \;=\; \big[\mathbb{E} \log p(Y \mid X) - \mathbb{E} \log p_Y(Y)\big] \;-\; \big[\mathbb{E} \log p(Y \mid X) - \mathbb{E} \log \widehat{f}_X(Y)\big].$$

The first bracket is $\mathrm{I}(Y; X)$ by definition. The second bracket is the expected log regret, which Proposition 2 evaluates as $\mathrm{I}(W; X \mid \kappa) + \mathbb{E}_\kappa \mathrm{KL}(G_\kappa \,\|\, \widehat{G}_\kappa)$. Substituting gives the display, and dropping the nonnegative KL term gives the bound. □

Remark 6 (Calibration does not exploit forecastability). Calibration moves $\widehat{G}_\kappa$ toward G_κ and so removes the KL term. That term depends only on the pooled law of the score, and a forecaster whose inputs carry no information about the outcome gains it just as much. The part of the gain that depends on the inputs is $\mathrm{I}(Y; X) - \mathrm{I}(W; X \mid \kappa)$, and the frozen score sets it. At a fixed score and stratification, calibration therefore exploits none of the forecastability. For example, suppose Y is independent of X but the location model varies with x. The residual then depends on X through the model alone, and a scheme with exact coverage scores below the unconditional law by $\mathrm{I}(R; X)$.

Remark 7 (Distributional, localized and folded-score schemes). The input can also return through the score. Distributional conformal prediction (Chernozhukov et al., 2021) scores by an estimated conditional rank. Its floor is therefore $\mathrm{I}(W;X)$ for that rank, which is zero when the estimated conditional law is the true one. It closes the gap through the estimate and then stops. Density-based, histogram, localized and quantile constructions (Izbicki et al., 2020, 2022; Sesia and Romano, 2021; Guan, 2023; Romano et al., 2019; Cauchois et al., 2021) likewise lower the floor by what their conditioning retains.

Localized calibration, which weights calibration scores by their nearness to the test input, lies outside the proposition, since its emitted law depends on the input beyond a fixed stratum. As the localization sharpens it can approach the conditional law and remove the floor. It is a change of stratification, and the proposition applies to each fixed one.

The folded absolute residual can be read as a symmetric predictive with density $h_{\mathrm{sym}}(z) = \frac{1}{2}(p_R(z) + p_R(-z))$. The same computation then adds $\mathrm{KL}(P_R \,\|\, H_{\mathrm{sym}}) \le \log 2$ for the discarded sign.

Remark 8 (Set loss, cutoff by cutoff). For a discrete outcome the identity also holds for prediction sets, one probability cutoff at a time. Zhang and Bates (2026) score a set Γ by $\ell_\lambda(Y,\Gamma) = |\Gamma| + \lambda^{-1}\mathbf{1}\{Y \notin \Gamma\}$ for a cutoff $\lambda \in (0,1]$, whose Bayes set is $\{y : p(y \,|\, X) \ge \lambda\}$, and define from it a conformal mutual information I_λ and a divergence D_λ. Let the score $A(x,\cdot)$ be one-to-one on the finite outcome set, so a set of scores pulls back to a set of outcomes of the same size and the same coverage. A frozen scheme at cutoff λ uses the pulled-back set $\{y : \widehat{g}_\kappa(A(x,y)) \ge \lambda\}$. Its excess set loss over the oracle is then

$$I_\lambda(W;X \,|\, \kappa) \;+\; \mathbb{E}_\kappa D_\lambda\big(G_\kappa \,\|\, \widehat{G}_\kappa\big)$$

at every cutoff, and at a fixed cutoff calibration moves only the second term.

Integrating over λ returns Proposition 2: $\int_0^1 I_\lambda \, d\lambda$ is the Shannon information (Zhang and Bates, 2026, Theorem 1), and $\int_0^1 D_\lambda \, d\lambda = \mathrm{KL}(G_\kappa \,\|\, \widehat{G}_\kappa)$ by the same calculation, in the tradition of Schervish (1989). So the log-score floor is the integral over cutoffs of what the frozen sets lose in size and coverage at each cutoff. The cutoff is not the nominal coverage level. A conformal set at level $1-\alpha$ corresponds to a data-dependent cutoff, and sees one slice of the integral. The identities, including an example where the outcome carries information about the input but the score does not, are checked in `check_setloss.py`.

The floor measures how the law of the score varies with the input. Whether an input is typical or unusual under its own marginal law does not enter, and a rare input with ordinary noise adds nothing to $\mathrm{I}(W;X \,|\, \kappa)$. Marginal coverage also does not say which inputs are over- or under-covered. We use *coverage* for the property of a set and *calibration* for the distributional property of a predictive.

4 The cost of stopping on 17,364 economic series

To measure the cost of stopping we need a forecaster in which the repaired map of Section 2 can sit at more than one position. We build one from the grammar of Table 1 and run it on 17,364 economic series. Stopping at the pooled law costs about 0.4 nats per observation one step ahead, composing a model after the map recovers a third of that, and interval coverage does not register the loss.

4.1 Moving the pooled law within a forecasting chain

Each chain in the grammar removes one kind of structure and passes its residual on. A terminal leaf models the law of whatever residual remains, and the predictive in the original coordinate is the leaf's law pushed back through the inverse chain. The base forecaster, `laplace`, runs roughly sixty candidate chains in parallel, combines their predictive densities with likelihood weights updated every observation, splices generalized Pareto tails onto the extremes, and scores its own probability integral transforms online. It ranks first in the public benchmark extended here, and its own terminal leaf is tuned to CRPS.

We compare two arms. Both use the same object, the kernel-smoothed pooled empirical law of recent residuals, but they are two forecasting architectures, not one map with and without a model after it.

stop. The base forecaster runs unchanged. Its predictive keeps its location, and its shape is replaced by the pooled law of the forecaster's own recent errors. That is the quoted forecast, and nothing is fitted to it afterwards. It is the online counterpart of the schemes of Proposition 2. Its location model and residual window evolve, so the proposition does not identify its deficit as a fixed floor.

compose-after. The pooled law moves inside the forecaster. In each candidate chain c it replaces the terminal leaf, so it describes that chain's own residual stream R_t^c instead of the final error. Write $\widehat{G}_t^c$ for that law and $(T_t^c)_{\#}^{-1}$ for pushing a law back through chain c's inverse transforms. The arm's predictive is

$$\widehat{F}_t \;=\; \sum_c w_t^c \left(T_t^c\right)_{\#}^{-1} \widehat{G}_t^c,$$

where the weight w_t^c is updated every observation from the density chain c assigned to the outcome.

The mixture is therefore a model fitted to the output of the empirical maps. Because every chain ends in an empirical map, the weights choose between chains and never between the empirical map and a parametric leaf. This arm has no coverage certificate.

We compare the arms by out-of-sample log likelihood, so each must give the realized outcome a usable density, including outcomes beyond the range of past residuals, where an empirical law, even kernel-smoothed, has almost none. The base forecaster and both arms therefore share one further step, a generalized Pareto splice that replaces the density beyond a fixed tail region. The pooled law is built from a window of recent residuals, and a stale window would lose likelihood for reasons unrelated to stopping. Both arms therefore run with flat and with recency-weighted windows (exponential weights, half-life 150 observations).

4.2 Series, tiers and the null control

The corpus is FRED economic change series (differences of rates, production, employment and similar), non-price, with two tiers. The *breadth* tier uses 256 observations of history per forecast origin and every qualifying series: 17,364. The *deep* tier uses 1,024 observations and series with at least 1,100, so the 400-residual window is a strict subset of the history it could have used: about 6,400 series.

Each series has twelve rolling origins, and we forecast at horizons 1, 6 and 12. We score each forecast by its log density, in nats. A deficit of 0.15 nats means the realized outcomes receive $e^{0.15} \approx 1.16$ times less probability per observation. We also compute CRPS, the continuous ranked probability score, which is in the units of the data and which most conformal papers report. Because origins within a series overlap, all intervals are bootstrap over series, not rows.

We floor the log density at -25 per observation. The floor is not a proper score where it binds, and it binds more often on the stopped arm (1.78% of rows) than on the base (0.62%, breadth tier, paired). Tightening the floor stepwise from -25 to -10 shrinks the raw stop deficit at horizon 1 monotonically from 0.51 to 0.33 nats, so over that range clipping works against the reported gap. The floor is also not invariant to the units of a series.

Part of what a constrained arm loses is the cost of estimating an empirical law from a finite window. To gauge it we run the identical protocol, at matched context, on a control corpus of synthetic iid draws (Gaussian, Student-t_3, Laplace). These series have no state to condition on, so what a constrained arm loses on them is estimation overhead alone: the price of a 41-node empirical law instead of a parametric scale. We subtract that overhead from every headline number below and call the result *null-control-adjusted*.

The adjustment is a gauge, not an exact correction. The estimation term $\mathrm{KL}(G \,\|\, \widehat{G})$ depends on G, and three simple iid laws need not match economic residuals. The base forecaster also adapts, so even on iid data its residuals are not exactly state-independent. Without the adjustment the breadth-tier deficits are 0.51, 0.61 and 0.59 nats at horizons 1, 6 and 12 (paired, 208,368 rows per horizon). The adjustment removes about a fifth of that.

4.3 Cost of stopping and the share recovered by composing

On the 17,364 breadth-tier series, the null-control-adjusted log score deficits against the unconstrained base are as follows.

	h=1	h=6	h=12
stop, total cost	0.397	0.475	0.462
of which pooled shape	0.267	0.315	0.275
of which position	0.131	0.160	0.187

Raw per-arm deficits carry series-bootstrap 95% intervals of about ± 0.02 throughout. Composing recovers 33%, 34% and 40% of the cost at the three horizons. The deep tier gives raw stop costs of 0.34 to 0.40 nats with 40 to 55 percent recovered by the interior placement.

The deep-tier fraction is not a pure position effect. Between tiers three things change at once: the context grows from 256 to 1,024 observations, the series panels differ, and the 400-residual window truncates a 1,024-observation history while holding all of a 256-observation one. A stopped map need not use a 400-observation window, since it may retain every residual it has ever seen. A same-series, same-context ablation would separate these effects, with stop arms at windows 250, 400, 750 and all history and composed arms matched window for window on the ctx-1024 panel. Until it runs, we do not know whether the cost of stopping grows with how much the pooled window discards.

The arms are close on the median series. The mean cost is carried by a minority of series where stopping hurts badly.

Adaptivity does not substitute for position. Recency-weighting the stop arm's window recovers 0.05 to 0.08 nats on the breadth tier, against 0.14 to 0.27 for moving the map's position,

raw in both cases. In the frozen setting of Proposition 2 level adaptation acts on the pooled scores alone and cannot touch the floor, and the online result is consistent with that.

On CRPS the tiers disagree. On the deep tier the stopped arm degrades CRPS and the composed arm improves it, while on the breadth tier both positions sit within about 1% of the base. We do not use CRPS in any summary.

4.4 Empirical coverage of the stopped and composed arms

Conformal practice monitors coverage, so we measure it too. The series are economic time series and we assert no exchangeability, so this is empirical coverage and not the finite-sample certificate of Section 1. The intervals are the 0.1 and 0.9 quantiles of the kernel-smoothed predictive, which need not coincide with the raw conformal order-statistic endpoints. The table therefore tests whether watching interval coverage would reveal the log score loss. It uses the breadth tier and paired rows, and the nominal coverage is 0.800.

arm	coverage of the 0.1–0.9 interval
unconstrained	0.870
stop	0.784
stop, weighted	0.799
compose-after	0.797

Every conformal arm sits nearer the nominal level than the unconstrained base, which overcovers at 0.870, so on coverage the conformal arms do better. On the log score the stopped arm loses 0.40 to 0.48 nats. Coverage tracks the rank of the outcome in the predictive, and the floor tracks conditional information the frozen score discarded. The two move independently, so monitoring coverage alone does not reveal the loss.

5 Discussion

Well known results place limitations on the conformal program. Exact distribution-free conditional coverage is impossible in nonatomic settings (Lei and Wasserman, 2014; Barber et al., 2021). Weighting calibration points unequally costs coverage in proportion to the total-variation distance between their residual laws (Barber et al., 2023). A guarantee held at a fixed point rather than on average inflates the threshold for classical testing-power reasons (Li, 1989; Low, 1997; Cai and Low, 2004). They are reasons to pool the scores, not reasons to stop at the pooled law.

That calibration alone does not establish forecasting skill is older than conformal prediction. Calibration in the sense of Dawid (1982) can be achieved, approximately and asymptotically, by randomized forecasts against any sequence of outcomes (Foster and Vohra, 1998). In a finite-horizon prequential setting, an uninformed randomized forecaster can pass, with high probability, any given test that accepts truthful forecasts whatever the data-generating law (Sandroni, 2003). Fishelson and Mohri (2026) calibrate forecasts of many outcomes at once using nothing but past outcomes. The identity of Section 3 prices the same fact for conformal schemes, as the forecastability that calibration leaves unexploited.

Training a neural network stops at an epoch chosen on validation evidence. The conformal recipe stops at its empirical map with no such evidence, and the census of Appendix A shows the stop is near universal, so we call it *cultural*. One likely reason is that the certificate belongs

to the stop. Coverage holds at the empirical map, so work that continues past it gives up the property the field measures itself by, and the field's effort has gone instead into better weights, strata and levels.

The raw one-step deficit of 0.51 nats means that the stopped forecast gives the realized outcome, on geometric average, about 60 percent of the density the unconstrained forecast gives it, and the coverage that conformal practice monitors registers none of this.

We call the stop a permanent structural information sacrifice, and both adjectives are meant literally. The sacrifice is structural because the score and the strata fix it before any calibration sample is drawn. It is permanent because nothing applied to the pooled scores, neither a level schedule nor a larger calibration sample, recovers it. The remedies lie elsewhere: upstream, in a score that leaves less of the input in the residual, or downstream, in a model fitted to the map's output, which is what the composed arm does.

The composed arm has no coverage certificate, but one can be added back by a second empirical map at the end of the composed chain, calibrated on fresh scores. Its floor is then the information left in the composed residual. That is smaller than the base residual's only if composing removed residual dependence, which a better log score does not by itself guarantee. A certificate can therefore go anywhere in a chain, like any transform in Table 1, and the model around it can be anything, a Bayesian posterior predictive included.

Limitations. The prices are measured on one base forecaster and one economic corpus. The floor proposition covers invertible signed scores, and the absolute residual only in the symmetric reading of Remark 7. Quantile scores need their own treatment.

A A census of the conformal literature

We ask what published conformal methods do with the fence-post map (1), which we also call the *empirical map*. A method *stops* if the output of the map, its quantiles, its p-values or its intervals, is the answer. It *composes* if it keeps the map and fits a model to its output, as in Section 2.

Methods that neither stop nor compose fall into three further classes. A method *replaces* if it substitutes a conditional estimate for the pooled quantile, as SPCI fits a quantile regression to recent residuals (Xu and Xie, 2023). It *tunes* if it keeps the map and adapts the level online, as adaptive conformal inference does (Gibbs and Candès, 2021). It *consumes* if it feeds the output to another procedure, such as training on pseudo-labels it has screened or selecting among candidates.

We classified the literature instead of relying on an impression of it. We took every arXiv paper with "conformal prediction" in the title as of 20 August 2026, 686 records. Six concern molecular or protein conformation and were matched by the title search in error, which leaves the remaining 680 papers.

Language-model readers (Claude Sonnet) coded each paper against a written rubric, read its body in full, and recorded the sentence that places it.[1]

[1]Two readers, each blind to the other, read the body of every paper from its first line to its reference list. The boundaries between stopping, replacing and consuming involve judgment, and the two readers disagree on 24 of the 680 papers. We read the method sections of every paper on which they disagree, and of every paper either of them coded as composing, and settled its label. An earlier pass that read only the abstract, the introduction and the method passages found the same two composing papers. Four papers have no PDF on arXiv and were coded

Of the 680 papers, 454 stop, 77 consume, 50 replace, 40 tune, and 57 propose no method. The two that compose do so narrowly. Messoudi and Destercke (2021) and Park et al. (2024) fit a copula to the outputs of per-target empirical maps, but use it only to set each target's quantile level and still return a box of conformal intervals. No paper models the rank stream of a single outcome or carries a fitted law back as a forecast. Lardy and Pérez-Ortiz (2024) come closest, fitting densities to conformal ranks, but to drive a test martingale.

Distributional conformal prediction (Chernozhukov et al., 2021) is the nearest single-outcome case. It conditions *before* the map, and its answer is the empirical quantile of the resulting ranks. A paper would count as composing if it kept the empirical map, applied it to scores, and fitted a model on the transformed stream. Reweighting the empirical law, conditioning upstream and adapting the level do not count.

The 77 papers that consume the conformal output use prediction sets or conformal p-values as inputs to selection, active learning, planning and similar procedures. Using the sets as data has its own limit. The law of the prediction sets does not identify a classifier's confusion matrix once a set of size two or more occurs, and a conformal correlation matrix built from set membership inherits the failure (Cotton, 2026c).

As an independent check, we coded by hand a random sample of 100 papers, drawn with a fixed seed from a wider archive of conformal prediction papers, without reference to any other labels. We find that 93 of the 100 treat the conformal step as terminal. The remaining 7 are unclear because they pass its output downstream as a filter or a feature. None fits a further law on the transformed scale and carries it back as a forecast. Sixty-two of the 100 are in the census, and the two codings agree on 61 of them.

The argument does not depend on the exact count. The count is reported because the uniformity is nearly complete, and that uniformity is itself a finding about the field. Outside conformal prediction the composing move is routine. The normal quantile transform in hydrological forecasting (Bogner et al., 2012), copula time-series models (Chen and Fan, 2006), and autoregressive transformation models (Rügamer et al., 2023) all apply an empirical marginal map and keep modelling in the transformed coordinates.

The readers also recorded how each method pools calibration scores. Of the 623 papers that propose a method, 452 pool all scores together, 77 pool within fixed strata, and 94 let the weights depend on the test point, through kernel localization, covariate-shift likelihood ratios or a sampling design.

B The repaired map

The piecewise-linear repair interpolates the pairs $(r_{(i)}, z_{(i)})$ and extends beyond the edge knots linearly in (r, z). Linearity in the probit coordinate is a Gaussian tail in the score, with scale set by the spacing of the extreme knots. A monotone C^1 cubic through the same knots gives a smooth repair. Both are strictly increasing bijections of the real line (Cotton, 2026b). A Gaussian-kernel estimate of the distribution function, which the experiments of Section 4 use, does not in general pass through the knots, so the argument below covers it only to within its distance from them. Section 4.4 accordingly reports empirical coverage.

The certificate survives any strictly increasing m_n through the knots. Let $R_1, \ldots, R_{n+1}$ be

from their abstracts. The rubric and the per-paper labels of both readers, with their quoted evidence, are in the repository.

exchangeable and almost surely distinct. Monotonicity gives

$$\{m_n(R_{n+1}) \le z_{(i)}\} \;=\; \{R_{n+1} \le r_{(i)}\}.$$

By exchangeability the rank of R_{n+1} among all $n+1$ scores is uniform on $\{1,\dots,n+1\}$, and $R_{n+1} \le r_{(i)}$ holds for exactly the i lowest ranks, so

$$\Pr\{m_n(R_{n+1}) \le z_{(i)}\} \;=\; \frac{i}{n+1}, \qquad i = 1,\dots,n.$$

Write $U = \Phi(m_n(R_{n+1}))$. For u between consecutive grid levels $i/(n+1)$ and $(i+1)/(n+1)$, the display and monotonicity of the distribution function give

$$\frac{i}{n+1} \;\le\; \Pr\{U \le u\} \;\le\; \frac{i+1}{n+1},$$

so $\Pr\{U \le u\}$ and u lie in the same cell of width $1/(n+1)$. The edge cells are the same argument with endpoints 0 and 1. Hence $\sup_u |\Pr\{U \le u\} - u| \le 1/(n+1)$, whatever the interpolation, the tails and the law of the scores.

C Proof of Proposition 2

Fix the fold, so (A, κ) is deterministic. Given $X = x$, change of variables gives the true density $p(y \mid x) = p_W(A(x,y) \mid x)\, A'(x,y)$, and the conformal density is $\widehat{f}_x(y) = \widehat{g}_{\kappa(x)}(A(x,y))\, A'(x,y)$ with $\widehat{g}_\kappa$ the density of the absolutely continuous estimate $\widehat{G}_\kappa$. The Jacobian cancels in the log ratio, so

$$\mathrm{KL}\big(P(\cdot \mid x) \,\|\, \widehat{F}_x\big) = \mathrm{KL}\big(P_W(\cdot \mid x) \,\|\, \widehat{G}_{\kappa(x)}\big) = \mathrm{KL}\big(P_W(\cdot \mid x) \,\|\, G_{\kappa(x)}\big) + \mathbb{E}_{W \sim P_W(\cdot \mid x)} \log \frac{g_{\kappa(x)}(W)}{\widehat{g}_{\kappa(x)}(W)}.$$

Averaging over X, the first term is $\mathbb{E}_X \,\mathrm{KL}(P_W(\cdot \mid X) \,\|\, P_W(\cdot \mid \kappa(X))) = \mathrm{I}(W; X \mid \kappa)$ by definition. Given $\kappa(X) = k$ the score has law G_k, so the second term averages to

$$\mathbb{E}_\kappa\, \mathbb{E}_{W \sim G_\kappa} \log \frac{g_\kappa(W)}{\widehat{g}_\kappa(W)} \;=\; \mathbb{E}_\kappa \mathrm{KL}\big(G_\kappa \,\|\, \widehat{G}_\kappa\big) \;\ge\; 0.$$

Every calibration operation is a function of the pooled scores alone, hence acts only on $\widehat{G}_\kappa$ and the second term. □

D Reproduction

Every constant regenerates from a tidy store keyed `(task, source, dataset, uid, model, horizon, repr, metric, value)`; the script `constants.py` in the source directory performs the regeneration and fails if the manuscript disagrees with the store. Breadth tier rows are tagged `forecast:CONFB:ctx256:.../m6+hash`, deep tier `CONF:ctx1024`, controls `CONFNB`/`CONFNULL`; the hash covers the harness sources, the tracked diff, and the compiled core.

The census rubric and the per-paper labels, with their quoted evidence, are in `census/` in the source directory, and the papers themselves are on arXiv. The hand-coded sample is `survey100.csv`. The exact checks of Section 3 are `check_gap.py` and, for Remark 8, `check_setloss.py`, in the source directory. Figure 1 and its value of $\mathrm{I}(R; X)$ come from `fig_gap.py`. Study declarations and the fill protocol are in the `skaters` repository (`surrogate/bench/studies.py` `skills/run-a-study.md`).

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